035: Where movement pays — the boundary of profitability in a resource-limited pond

- **Type:** research article (short format, one result)
- **Version:** v1.0
- **Date:** 2026-11-07
- **Status:** reviewed by the lead (07.11); ready for deposit once the author says so
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- **Previous article of the series:** 032
- **Artifacts:** `morphogenesis/experiments/E09_pond/s49_e1_density_axis.json`

Abstract

The previous articles of this series established that a cell's behaviour pays or does not pay depending on the environment, but they measured single points. This article maps the surface. We ran 240 simulations across four colony densities and three resource-recovery rates, ten seeds each, and measured the difference between a cell that steps towards the richest neighbour and a cell that does not move.

The boundary exists, and it runs along the **recovery rate**. With medium or fast recovery, movement pays at every density tested (+0.028 to +0.076). With slow recovery the sign flips at the two highest densities (−0.022 and −0.030). The optimum is at a density of 128 cells on 1024 (12.5 %), not at the lowest density tested (3.1 %): with too few cells nobody finds a patch, and at 53.7 % the competition eats the gain.

This explains an earlier result of the series: the working density of 53.7 % sits beyond the boundary when recovery is slow, which is why greedy movement was found to be harmful there.

The line this article continues

The previous article of this series (032) found that greedy movement was *harmful*: at the working density of 53.7 % with slow resource recovery, a cell that stepped towards its richest neighbour did worse than a cell that stayed put (gap −0.174). That result was measured at a single point, so it could not say whether the harm came from the rule itself or from the place on a larger map where the experiment happened to sit. This article measures the map: four colony densities crossed with three resource-recovery rates, ten seeds each, to find where the sign of the gap changes and why.

Keywords: agent-based model, foraging, density dependence, resource renewal, boundary of profitability

How this article was made. It was written by an AI agent working inside the project's repository, from artifacts the agent also produced, with periodic feedback from the human operator who owns the project. The operator set the frame and the rules and reviewed drafts; the agent ran the experiments, wrote the text, built the figures, wrote the machine check of every printed number, and prepared the

deposit. Where the article says "we", it means this collaboration. The artifacts and the check are in the deposit, so every claim can be re-derived without trusting either of us.

The line this article continues

Last time. *What decides whether behaviour pays: movement, not the signal it follows* (article 032, v1.2) showed that the sign of the greedy-movement advantage depends on the resource recovery rate — it loses under slow recovery and wins under fast — and that the dominant variable is how much a cell moves, not what it compares with. *Directionality works but does not scale* (article 031, Zenodo DOI 10.5281/zenodo.24012536) had already found that the gain decays as the rich zone fills.

What we decided to check. Both results were single points: one density, one recovery rate, one mobility. This article maps the surface — four densities × three recovery rates, ten seeds each, 240 runs — and asks where movement stops paying.

1. The question

The series has measured the value of movement at single points: one density, one recovery rate, one mobility. That answers "does it pay here" but not "where does it stop paying". The literature calls the second question the stronger one — a surface rather than a point — and it is what this article measures.

2. Methods

The world. A 32×32 grid with resource in the cells, organisms with energy, proportional income, death at zero energy, reproduction on accumulated mass. Three rich patches; the colony is seeded on one of them.

The two arms. `baseline` never moves; `greedy` steps to the free neighbour with the most resource when its own resource is below its running prediction. Everything else is identical.

The grid. Four densities (32, 128, 320, 550 cells on 1024, i.e. 3.1 %, 12.5 %, 31.2 % and 53.7 %) × three recovery rates (`resource_regen` 0.01, 0.02, 0.04) × ten seeds × two arms = **240 runs** of 600 steps.

The metric. The favourable share — the fraction of living cells standing on the top half of cells by current resource — and the gap between the arms. R1 (mean deaths ≥ 10 and mean living ≥ 10) holds in **240 of 240** runs.

3. Results

recovery	density 3.1 %	12.5 %	31.2 %	53.7 %
slow (0.01)	−0.008	**+0.032**	−0.022	**−0.030**
medium (0.02)	+0.047	**+0.063**	+0.035	+0.031
fast (0.04)	+0.054	**+0.076**	+0.038	+0.028

Three things follow.

1. **The boundary exists and it runs along the recovery rate.** With medium or fast recovery movement pays

at every density; with slow recovery the sign flips at 31.2 % and 53.7 %.

2. **The optimum is at 12.5 %, not at the lowest density.** At 3.1 % there are too few cells to find a patch; at 53.7 % the competition consumes the gain. The optimum is also the only cell of the table where every seed agrees: 10 of 10 positive at 12.5 % with fast recovery, against 4 of 10 at 31.2 % and 1 of 10 at 53.7 % under slow recovery. The two negative cells are therefore a shift of the distribution, not a reversal in every run. Under medium and fast recovery every cell is positive in 10 of 10 seeds.

3. **The highest density tested is the worst case for movement**, and it is the density the earlier experiments of this series used.

4. Discussion

What this explains. Article 032 reported that greedy movement was *harmful* under slow recovery (−0.174). This map shows why: that experiment ran at 53.7 % density with slow recovery, which is exactly the cell of the table where the gap is negative (−0.030). The earlier result was not a property of the rule but a position on this surface.

What cannot be concluded. We do not show that the boundary is universal — it is measured in one architecture, at one mobility (one step per tick) and with ten seeds. The literature recommends 20–30 seeds for a boundary; this is a pilot with enough seeds to see the shape, not to fix the crossing point.

Relation to the literature. The frame — that the value of movement depends on how much resource there is and how fast it returns — is not new; it is the cost–benefit frame of Mathew & Sasidevan (2019), the first source of article 031. What is measured here is the **concrete form** of the boundary in this architecture: the position of the optimum (12.5 %) and the fact that it disappears when recovery is fast.

5. What this does not show

- It does not give a formula for the boundary; the surface is measured, not derived.
- It does not test mobility: the whole map is at one step per tick. The mobility axis is the obvious next one.
- It does not show that 12.5 % is the optimum in general — only that it is the optimum among the four densities tested.

6. Where we move next

- **Add the mobility axis**: the same map at three mobilities, to see whether mobility shifts the boundary or only the height of the gain.
- **Refine the peak**: densities 64, 96, 128, 160, 192 with 20 seeds, to locate the optimum rather than bracket it.

7. Reproducibility

The single artifact `s49_e1_density_axis.json` holds all 240 rows; the script is `morphogenesis/scripts/s41_e1_phase_diagram.py`. Runs were on CPU with one thread. R1 is true in every run.

Authorship and automation

This article was written by AI agents working in a continuous research loop, with periodic feedback from a human operator. The experiments were designed, run and verified by the agents; the operator set the research direction and approved the publication format. Every number in the text is machine-checked against the deposited artifacts by `scripts/check_article_035_numbers.py` (21 checks).